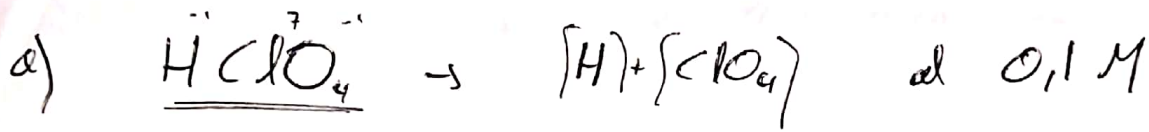
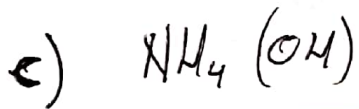
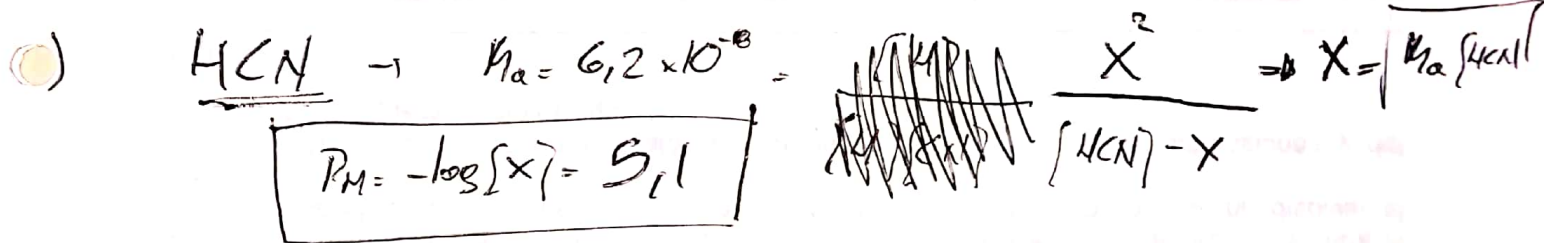
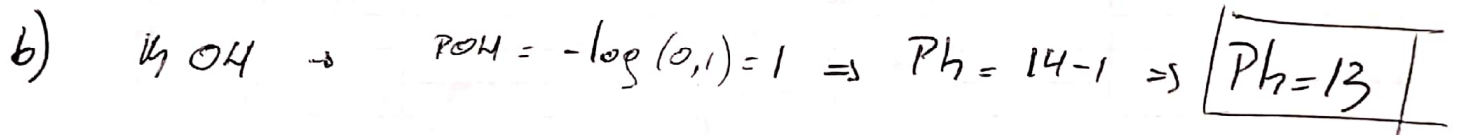


Problema N° 1

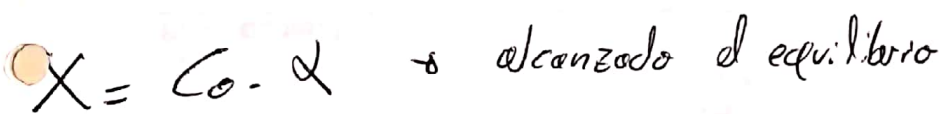


$$\text{pH} = -\log [0,1] = \boxed{1 = \text{pH}}$$



$$\log_{10} [\text{H}] = \text{pH} \Rightarrow 10^{\text{pH}} = [$$

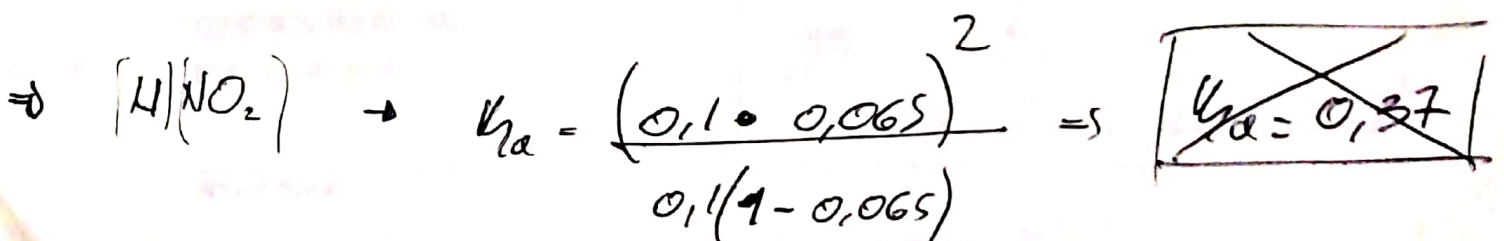
Problema N° 5



$$K_a = \frac{(C_0 \cdot \alpha)^2}{C_0(1 - \alpha)}$$

Si $\alpha = 1 \rightarrow$ totalmente "disociado"

$\alpha = 0 \rightarrow$ No se disocia $\rightarrow X \rightarrow$ chiquito



caso N° 2

HCl (FuenteFuote) $\rightarrow$ en 100 mL solución $\rightarrow$ 3,65 g HCl

1 mol HCl $\rightarrow$ 36,5 g

~~0,054~~ = X $\rightarrow$ 3,65
0,054

$$\Rightarrow \text{Ph} = -\log \left(\frac{0,1}{0,01} \right) \Rightarrow \boxed{\text{Ph} = 5 \times 10^{-3}}$$

$$M = \frac{n}{L} = \frac{0,0098}{0,1 \text{ (L)}} = 0,098 \text{ M}$$

$$\text{Ph} = 12 \rightarrow \text{POH} = 2 \Rightarrow -\log_{10} [\text{OH}] = 2$$

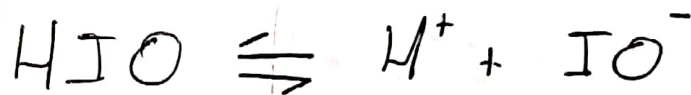
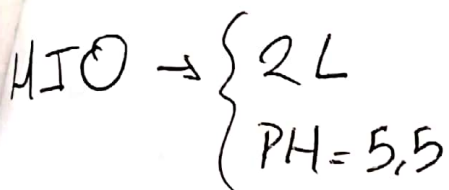
Problema N° 8

$$\boxed{[\text{OH}] = 0,01}$$

$$10^2 = 100 \Rightarrow -\log_{10} \frac{[\text{H}^+]}{100} = \text{Ph}$$

$$\boxed{-\text{Ph} = 10 = [\text{H}^+]}$$

Problema N° 7



• $n_{\text{HIO}}?$

$t=0$ C_0 $-$ $-$

• $K_a = 2,3 \times 10^{-11}$

$t=eq$ $C_0 - x$ x x

$$K_a = \frac{[\text{H}][\text{IO}]}{[\text{HIO}]}$$

$$\frac{x \cdot x}{(C_0 - x)} = 2,3 \times 10^{-11} \quad (1)$$

Como $K_a \ll 1$ se favorece los reactivos $\Rightarrow$ x muy chiquitas

$$\text{Ademas} \Rightarrow \text{pH} = -\log[\text{H}^+] \Rightarrow [\text{H}^+] = x$$

$$\Rightarrow \text{pH} = -\log(x) \Rightarrow \text{pH} = 5,5$$

$$x = 10^{-5,5}$$

$$\Rightarrow x = 3,16 \times 10^{-6} \quad (2)$$

reemplazando en (1)

$$\frac{x^2}{C_0 - x} = 2,3 \times 10^{-11}$$

$$\Rightarrow C_0 = 0,43 \frac{\text{mol}}{\text{L}}$$

Si en 1 L — 0,4 mol

2 L $\rightarrow$

$$n = 0,86 \text{ mol}$$

Porcentaje de disociación

$$\alpha = \frac{X}{C_0} \cdot 100 \Rightarrow \frac{3,16 \times 10^{-6}}{0,43} \times 100$$

$$\boxed{\alpha = 7,3 \times 10^{-4} \%}$$

↪ Se disocia muy poca cantidad

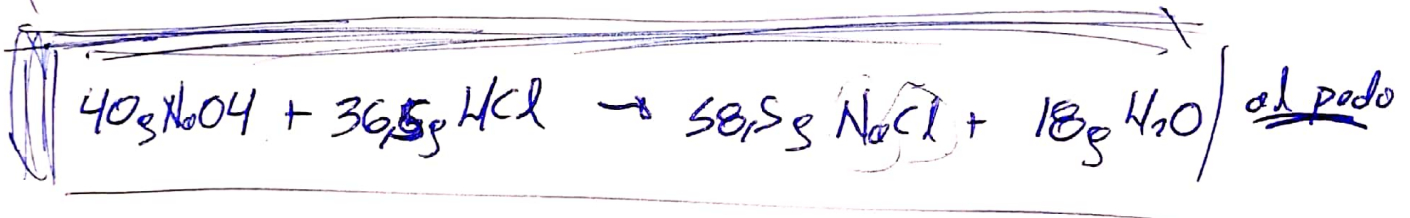
Exmo N° 9

PH

PH final?

25 ml NaOH a 0,16 M

50 ml HCl a 0,1 M



¿Cual es el exceso?

→ NaOH → En 1 l → 0,16 mol

0,025 l → X = 0,004 mol NaOH

HCl → En 1 l → 0,1 mol

0,05 l → X = 0,005 mol

de (1) 1 mol NaOH → 1 mol HCl

0,004 mol NaOH → X = 0,004 HCl

PL → NaOH

PE → HCl → 0,005 mol

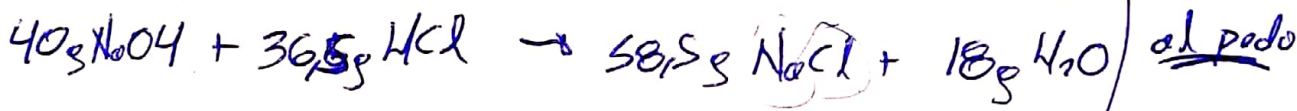
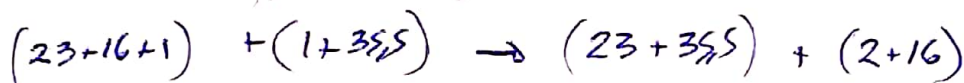
Nº 9

PH

PH final?

• 25 ml NaOH a 0,16 M

• 50 ml HCl a 0,1 M



¿Cual es el exceso?

→ NaOH → En 1 l → 0,16 mol

$$0,025 \text{ l} \rightarrow X = 0,004 \text{ mol NaOH}$$

HCl → En 1 l → 0,1 mol

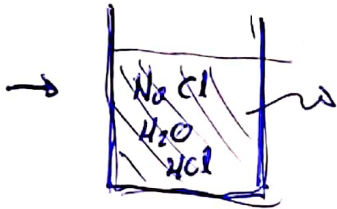
$$0,05 \text{ l} \rightarrow X = 0,005 \text{ mol}$$

de (1) 1 mol NaOH → 1 mol HCl

0,004 mol NaOH → X = 0,004 HCl

PL-NaOH

PE-HCl → 0,001 mol



$$V = 25 + 50 \text{ ml} = 75 \text{ ml soln}$$

$$[H^+] = \frac{0,001 \text{ mol}}{0,075 \text{ l}} = 0,013$$

$$PH = -\log [0,013] \Rightarrow PH_{\text{Final}} = 1,88$$